Experiment 7c: Eddy Current Ring

PURPOSE

Learning goal
Use changing magnetic flux, the conducting path, induced current, magnetic force, motion, and heating to build one causal explanation.

Quick launch

- Open experiment_7c_eddy_current_teacher_ready_v13.html in a modern browser. No server, internet connection, or external font is required.
- Project the page or let students use one device per pair. Use the four investigation tabs from left to right.
- Changing supply, ring type, or turn count deliberately resets the run and returns power to OFF, so each comparison begins from the same baseline.
- Use Slow motion for DC switching events and Reasoning chain only after students have committed to an explanation.

Suggested 18-minute lesson flow

Time	Phase	Teacher move	Student evidence
0-3 min	Predict	Keep power off. Ask what must change before a current can be induced.	Written prediction using field or flux.
3-8 min	DC transient	Run ON, wait for steady DC, then run OFF. Pause after each event.	Signs of e.m.f., ring current, and force; ring height.
8-12 min	Closed path	Use AC at 1100 turns. Compare complete and slit rings.	Current path, sustained lift, and heating.
12-16 min	AC and turns	Test 600, 1100, and 1600 turns for equal observation times.	Qualitative rank from current, force, and settled height.
16-18 min	Explain	Reveal the reasoning chain; require a full causal sentence.	Changing flux -> e.m.f. -> current -> force -> motion and heat.

Teacher language to keep consistent

Say changing magnetic flux, not only changing magnetic field. State that the current, e.m.f., and force readouts are signed. Positive force is upward; negative force is downward. The height-force graph scales force for visual comparison; the numerical force card is the value in newtons.

Expected evidence and explanation

OBSERVATION MATRIX

Test	What students should observe	Physics interpretation
DC switch-on Complete ring	A brief e.m.f. and ring current; an upward force pulse; the ring rises briefly.	Coil current and flux increase. Lenz's law gives a ring current whose interaction produces brief repulsion.
Steady DC	Coil current remains, but e.m.f. and ring current decay; lift disappears and the ring returns.	Flux is nearly constant, so d(Phi)/dt is nearly zero. A magnetic field alone does not maintain induction.
DC switch-off Complete ring	The e.m.f. and ring-current signs reverse; the force transient is downward/attractive.	Collapsing flux is still changing flux. The induced current reverses to oppose the decrease.
AC Complete ring	Ring current alternates, heating continues, and the ring reaches a sustained height.	AC keeps d(Phi)/dt non-zero. The mechanical response follows the cycle-averaged upward effect.
AC Slit ring	The main ring current is zero in the model; lift and heating are suppressed.	The slit breaks the dominant closed conducting path.
AC turns 600 / 1100 / 1600	The model gives a larger qualitative steady response at the higher turn settings.	Use as a controlled model comparison, not as a calibrated prediction for every real coil.

Causal chain represented by the simulator

1 Changing flux	2 Induced e.m.f.	3 Ring current	4 Magnetic force	5 Motion + heat
$\mathrm{Phi} = M(z)\, I_c$	$e = -d\mathrm{Phi}/dt$	Closed path required	$F_z = I_c I_r dM/dz$	$m z'' = F_z - mg - b z'$ $P = I_r^2 R$

Model evidence from the packaged QA run

Check	Result
DC switch-on maximum centre height	22.50 mm from a 12.00 mm baseline
Steady DC induced e.m.f.	Approximately 0.000 mV after decay
DC switch-off minimum force	-2.02 N transient
AC complete ring at 1100 turns	23.13 mm after 2.0 s
AC slit ring	12.00 mm; main ring current = 0 A
AC qualitative turn comparison	600: 19.51 mm; 1100: 23.14 mm; 1600: 25.16 mm

These values verify internal behavior at the packaged settings; they are not calibration targets for a physical apparatus.

Assessment, misconceptions, and model scope

USE WITH JUDGEMENT

Three-item exit ticket

1. Explain why steady DC does not keep the complete ring lifted, even though a magnetic field remains.
2. Explain why the slit ring behaves differently under the same AC supply.
3. Use the sign of the force and current to explain the DC switch-off event.

Success criterion

A complete answer includes the full chain: changing flux -> induced e.m.f. -> closed-path current -> magnetic force -> motion and heating. It distinguishes switch-on, steady DC, and switch-off.

High-value prompts

- What is changing?
- Is the conducting path closed?
- Which sign changed?
- Is the force upward or downward?
- Which displayed value is actual, and which graph trace is scaled?

Common misconceptions to intercept

Field means current. Redirect attention to d(Phi)/dt.

Power OFF means no induction. A collapsing field gives a brief changing flux.

Negative force means no force. Negative is the chosen downward direction.

Aluminium alone is enough. The dominant eddy current also requires a closed path.

The two curves share one physical scale. The force trace is scaled; use the force card for newtons.

More turns always gives the same real-lab ranking. The three buttons are a qualitative controlled comparison in this model.

Model scope and limits

Represented	Deliberately simplified
Coupled primary-secondary lumped circuits; position-dependent mutual inductance; signed e.m.f., current, and vertical force; gravity and damping; I^2R heating; DC ramp and 50 Hz AC.	Rigid one-dimensional ring motion; slit ring treated as open for the dominant loop; cycle-smoothed force for mechanical motion; qualitative thermal display; no detailed field solution, skin effect, core saturation, contact mechanics, or apparatus calibration.

Troubleshooting

If the apparatus seems still, check whether the page says Pause or Resume, then press Reset. A setup change returns power to OFF by design. On a smaller screen, the controls appear before the apparatus; scroll down to the live evidence. The simulator is a single offline HTML file, so blocked web fonts cannot stop the animation.